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Equine Infectious Disease

Effect of Long-Term Freezing on Indirect Fluorescent Antibody Titers for the Diagnosis of Equine Protozoal Myeloencephalitis

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ABSTRACT

Background: Long-term freezing storage can alter the stability of proteins, thereby compromising accurate determination of indirect fluorescent antibody test (IFAT) titers that support the diagnosis of equine protozoal myeloencephalitis.

Objective: Assess the effect of long-term storage at −80°C on IFAT against *S. neurona* and *N. hughesi* in equine serum and cerebrospinal fluid (CSF).

Samples: Paired serum and CSF ($n = 46$), and serum only ($n = 25$) samples.

Methods: Prospective study of samples stored 6–12, 13–18, and 19–24 months. Comparing antibody titers across time points, McNemar and Wilcoxon rank tests were used for statistical analysis.

Results: After long-term freezing storage for 19–24 months, *N. hughesi* IFAT results for CSF were significantly different compared with their fresh-testing time point ($p = 0.04$), but a similar difference was not observed for serum samples. No statistical differences were observed for serum or CSF samples tested for *S. neurona* after 6–12, 13–18, or 19–24 months compared with their original fresh testing results. However, the serum to CSF ratio cutoff of ≤ 64 in support of disease changed in 41% (19 of 46) of paired samples.

Conclusions and Clinical Importance: Freezing samples at −80°C for more than 18 months might alter test results, leading to possible changes in serum to CSF ratio calculations and affecting the interpretation of results for clinical and research purposes.

1 | Introduction

Equine protozoal myeloencephalitis (EPM) is a progressive multifocal neurologic disease affecting horses in the Americas caused by protozoa [1, 2], most frequently *Sarcocystis neurona* [2] and less frequently *Neospora hughesi* [2, 3]. The definitive diagnosis of EPM can only be made postmortem, by confirming the presence of protozoal parasites in the central nervous system [2]. A clinical diagnosis of EPM can be suspected in horses displaying progressive asymmetric focal or multifocal neurological

Abbreviations: CSF, cerebrospinal fluid; EPM, equine protozoal myeloencephalitis; IFAT, indirect fluorescent antibody test.

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signs of spinal cord disease, multifocal neurogenic muscle atrophy, brain disease, or a combination of these [3]. Acute, subacute, and chronic manifestations can occur [3]; and the exclusion of other neurologic causes further increases the suspicion of EPM in endemic areas or in horses imported from endemic areas.

Confirming intrathecally-produced antibodies in cerebrospinal fluid (CSF) using validated immunodiagnostic tests on both serum and CSF, such as indirect fluorescent antibody test (IFAT) and *S. neurona* surface antigen (SnSAG 2/4/3) enzyme-linked immunosorbent assays (ELISAs), is supportive of EPM as the clinical diagnosis [3–5]. Co-infections with more than one protozoa (*S. neurona* and *N. hughesi*) have been reported [6, 7]. Molecular detection of *S. neurona* using rtPCR for the diagnosis of EPM has been investigated with variable results [8, 9].

Freezing is commonly used to facilitate transport and to store serum and CSF samples before performing IFAT to diagnose EPM and for research purposes [10]. Reported advantages of frozen storage include increased stability and decreased microbial growth [11]. Nevertheless, protein destabilization and aggregation can occur during low-temperature storage or freeze–thawing cycles [12]. Several studies in humans have evaluated the effect of freezing on the concentration of antibodies for different diseases and the storage of monoclonal antibodies [12–16]. The results reported by these studies are variable, with some reporting no effect and others reporting antibody instability resulting in either increased or decreased antibody titers [13–16].

It is important to evaluate the repeatability of IFAT from samples stored under freezing conditions because clinicians commonly rely on these results for diagnosis, follow-up, and research purposes. To our knowledge, no studies have evaluated the effect of long-term frozen storage on antibodies against pathogens associated with EPM, tested using IFAT, in both serum and CSF. Therefore, our objective was to assess the effect of long-term storage at −80°C on antibody titers against *S. neurona* and *N. hughesi* in serum and CSF of horses. We hypothesized that antibody titers for both pathogens would be affected by long-term storage at −80°C for more than 18 months.

2 | Materials and Methods

Our prospective study compared the IFAT results of 3 groups of serum and CSF samples stored in screw-top plastic cryovial tubes (Self-standing Cryovial, Genesee Scientific, San Diego, CA) with different times of freezing storage at −80°C. For each freezing storage time point group, the baseline antibody titers of the samples were compared with the retested IFAT conducted during the study.

2.1 | Sample Size Calculation

An alpha level cutoff of 0.05% and 80% power were considered significant. The sample size calculation indicated that a sample size of $n = 7$ samples per time point group should be sufficient to detect a one-step (2-fold) difference in antibody titers (equivalent to 1 $\log_2$ unit) between freezing storage time point groups. This minimum target sample size was exceeded by the actual sample sizes for *S. neurona* for each time point group but not for *N. hughesi* time point groups. Thus, more extensive statistical analyses for *N. hughesi* were not possible because of limited sample availability. The sample size calculation was conducted using R software (4.3.1).

2.2 | Serum and CSF Samples

Equine serum and CSF samples used for the study were collected, assayed in a blinded manner, analyzed, and stored at the Immunology Laboratory at the William R. Pritchard Veterinary Medical Teaching Hospital at UCD between January 2022 and July 2023. The samples received at the laboratory were sourced from our in-house hospital equine patients and from external patient submissions shipped on cold packs by referring veterinarians as instructed by our diagnostic laboratory. Antibodies from equine samples stored under refrigeration or at temperate room temperature have been reported to be stable for IFAT for at least 10 days [17]. Upon arrival, samples were processed for IFAT and stored at −80°C thereafter until retested at the designated time point. Therefore, all samples were thawed only once for the purpose of the study. The inclusion criteria consisted of samples previously confirmed to have antibodies, independent of titer, for *S. neurona*, *N. hughesi*, or both, detected using IFAT performed as previously described [18]. The initial antibody titers were considered the baseline titers for the storage time point comparison group. The study was approved by the University of California Davis Institutional Animal Care and Use Committee (IACUC, protocol #23833).

Total archived samples with detectable antibodies against *S. neurona* and *N. hughesi* included 46 paired serum and CSF and 25 serum only samples. Of the 71 serum samples, 17 had antibodies against both pathogens. Thirty-five paired serum and CSF and 19 serum only samples had detectable antibodies against *S. neurona*, and 11 paired serum and CSF and 6 serum samples had antibodies against *N. hughesi*. Two of the 46 paired samples had detectable antibodies for both pathogens in both serum and CSF. All samples were divided into 3 groups according to their freezing storage time: 6–12, 13–18, or 19–24 months of storage, respectively (Table 1).

2.3 | Indirect Fluoresent Antibody Testing Analysis

The samples were thawed once at room temperature (20°C–23.3°C), and IFAT was performed as previously described [18], using *S. neurona* merozoites and *N. hughesi* tachyzoites [19] as the test antigens, at the UCD One Health Institute Laboratory. The serum used was serially diluted using two-fold dilutions from 1:40 to 1:20480, and then 10 μL aliquots were added to each well on the antigen slides. A fluorescein-labeled, species-specific horse IgG secondary antibody (Jackson ImmunoResearch, West Grove, PA) diluted 1:100 was added to visualize parasites for titer determination. The endpoint titer was the last serum dilution showing distinct, whole-parasite fluorescence. The IFAT was performed on CSF samples using the same methods, with an initial starting dilution of 1:5, with endpoint titers up to 1:5260 [19]. All completed slides were read blindly by the same

observer to determine endpoint titers and ensure consistency of results. In addition, slides were read by a second reader to check for reproducibility. For comparing antibody titers when retesting samples, a difference of more than 2 dilutions (increased or decreased) was considered clinically relevant because of test reading variability [20]. To complement this information, an IFAT serum to CSF ratio for *S. neurona* and *N. hughesi* ≤ 64 was interpreted as intrathecal antibody production [8] and supportive of EPM as the clinical diagnosis.

2.4 | Statistical Analysis

Non-parametric statistical analyses were conducted using R software (4.3.1) and MedCalc (MedCalc Software Ltd., Belgium). The Wilcoxon signed rank test was used for comparing antibody titers across time points. The McNemar test was used for comparing paired proportions of serum to CSF ratios. For serum to CSF ratio sample calculations, a small value of 0.1 was added to negative (undetectable antibodies) IFAT data cells to make it possible to calculate the serum to CSF ratios as previously described [8]. Statistical significance was defined as $p < 0.05$. Inter-reader variability was compared to determine repeatability across two independent readers during the study.

3 | Results

3.1 | Indirect Fluorescent Antibody Testing for *S. Neurona*

Overall, no significant differences were found in antibody titers in serum or CSF for *S. neurona* between baseline and the 6–12, 13–18, or 19–24 month time point groups. No significant differences were found for serum to CSF ratios for these time points (see below). However, the calculated ratio to support or rule out disease changed in 13 of 35 paired samples when samples were retested after storage (Table 2). Of these 13 paired samples, 10 ratios previously interpreted as positive at baseline became negative upon retesting, and 3 negative samples became positive. Between-reader comparisons of antibody titer determination for *S. neurona* reproducibility showed that both readers agreed within 0 to 2 dilutions (accepted variation for the test) on 100% of the tested samples.

TABLE 1 | Serum and CSF samples for *S. neurona* and *N. hughesi* for each freezing storage time point group.

	Sarcocystis neurona			*Neospora hughesi*		
Samples	**Serum**	**CSF**	**Paired serum:CSF**	**Serum**	**CSF**	**Paired serum:CSF**
6–12 m	9	9	9	2	2	2
13–18 m	26	16	16	4	2	2
19–24 m	19	10	10	11	7	7
Total	54	35	35	17	11	11

TABLE 2 | Serum to CSF ratios.

Total paired samples = 46				
Altered interpretation of results = 19/46 (41%)				
***S. neurona* serum:CSF ratio**				
	Baseline to 6–12 m (*N* = 9)	**Baseline to 13–18 m (*N* = 16)**	**Baseline to 19–24 m (*N* = 10)**	**Total (*N* = 35)**
Unaltered	7	8	7	22
Altered (%)	2 (22%)	8 (50%)	3 (30%)	13 (37%)
≤ 64	1	2	0	3
> 64	1	6	3	10
***N. hughesi* serum:CSF ratio**				
	Baseline to 6–12 m (*N* = 2)	**Baseline to 13–18 m (*N* = 2)**	**Baseline to 19–24 m (*N* = 7)**	**Total (*N* = 11)**
Unaltered	0	1	4	5
Altered (%)	2 (100%)	1 (50%)	3 (43%)	6 (55%)
≤ 64	0	0	0	0
> 64	2	1	3	6

Note: Number of cases with paired samples (serum and CSF) for the calculation of ratios for *S. neurona* and *N. hughesi* presented here. *N* = Number of paired samples (serum and CSF) to calculate ratios; unaltered = calculated ratio did not change for the interpretation of results from baseline; altered = calculated ratio changed the interpretation of results from baseline; % = percentage of paired samples with altered ratios.

For the 6–12-month time point group, no significant differences were found in the antibody titers for *S. neurona* between baseline and this time point (serum, $p = 0.56$; CSF, $p = 0.89$). The serum to CSF ratios also did not show a significant difference as a group ($p = 1$; difference in proportion of positive samples and 95% confidence interval [CI] = 0 [−30 to 30.8]) between baseline and retesting time point. In a single set of paired samples that had a 3 dilutions difference in both serum and CSF, the serum to CSF ratio did not change from baseline, whereas the serum to CSF ratio changed to above or below the cutoff value in two paired samples (Table 2). In one set of these samples, the ratio decreased from 128 to 64 because the antibody titer in CSF increased by 1 dilution at retesting. The ratio in the second set of samples increased from 64 to 256 because the antibody titers in serum increased by 2 dilutions at retesting.

For the 13–18-month storage time point, no significant differences were found in antibody titers against *S. neurona* between baseline and this time point (serum, $p = 0.25$; CSF, $p = 0.34$). All 26 serum samples did not differ ($n = 8/26$) or were within 1 ($n = 12/26$; 8 increased, 4 decreased) to 2 ($n = 6/26$; 5 increased, 1 decreased) dilutions difference. Two of 16 CSF samples had more than 2 dilutions difference (decrease 3 and 5, respectively), whereas 14 samples did not differ ($n = 4/14$) or were within 2 dilutions (1 dilution, $n = 9/14$ [3 increased, 6 decreased]; 2 dilutions, $n = 1/14$ increased). Although no significant difference was found in serum to CSF ratio between baseline and this time point ($p = 0.29$; CI, −25 [−57.4, 7.4]), 8 of 16 changed from their baseline ratio to above ($n = 6/6$) or below ($n = 2/6$) the cutoff value (Table 2). This set of samples had dilution differences in serum ($n = 1/6$), CSF ($n = 2/6$), or both ($n = 3/6$). In two sets of samples on which the ratio increased > 64, differences included 3 and 5 dilutions decrease, respectively.

For the 19–24-month period, no significant difference was found in antibody titers against *S. neurona* between baseline and this time point (serum $p = 0.07$; CSF $p = 0.09$). Serum samples did not differ ($n = 5/19$) or were within 1 ($n = 9/19$; 6 increased, 3 decreased) to 2 ($n = 5/19$ decreased) dilution differences. Six of 10 CSF samples varied by 1 dilution (5 increased, 1 decreased), and 4 samples did not change. No significant difference was found in serum to CSF ratios as a group ($p = 0.25$; CI, −30 [−58.4, 1.6]). However, 3 of 10 serum to CSF ratios changed to > 64 (Table 2). Two of 3 ratio changes resulted from 1 dilution difference in serum exclusively, and the third one had dilution differences in both serum and CSF (2 in serum, 1 in CSF).

3.2 | Indirect Fluorescent Antibody Testing for *N. Hughesi*

Between-reader comparisons of IFA titer determination for *N. hughesi* reproducibility showed that both readers agreed within 0 to 2 dilutions (accepted variation for the test) on 100% of the tested samples. Because of the small number of available archived samples with detectable IFA against *N. hughesi* in the 6–12 month and 13–18 month time points, further data analysis, beyond the descriptive comparisons below, could not be performed. Six of 11 paired serum and CSF samples had ratios that changed, with all 6 being interpreted as positive for disease at baseline to negative upon retesting (Table 2).

For the *N. hughesi* samples retested at the 6–12 month time point, two paired samples (serum, CSF) with detectable antibodies were available. Serum antibody titers decreased from 1:1280 to 1:640 (1 dilution decrease), and in CSF from 20 to 5 (2 dilution decrease) in both horses. This change resulted in serum to CSF ratios increasing from 64 at baseline to 128 at retesting for both horses.

For the 13–18-month time point, 4 serum and 2 CSF samples (two paired) were available. Serum *N. hughesi* antibody titers decreased by 3 dilutions in 1 of 4 samples. Antibodies were undetectable at retesting in 1 of the 2 CSF samples (the baseline antibody titer for that sample was 10). By giving a value of 0.1 for this CSF sample for statistical purposes and calculation of the ratio, the ratio remained > 64. The second serum to CSF ratio changed to > 64 (from 32 to 256) as the result of dilution differences in serum (1 dilution increase) and CSF (2 dilution decrease).

For the 19–24 month time point, 11 serum and 7 CSF samples (7 paired) with detectable antibodies were available for *N. hughesi*. No significant difference was found between serum samples at baseline and this time point ($p = 0.07$). A significant difference was found between CSF samples at baseline and this time point ($p = 0.04$). In serum samples, 2 of 11 samples had a retesting antibody titer that differed by more than 2 dilutions (3 dilutions decrease) compared with baseline antibody titer. In CSF samples, 4 of 7 samples did not have detectable antibodies against *N. hughesi* upon retesting. The baseline antibody titers were 10 ($n = 3$ samples) and 40 ($n = 1$). No significant difference was found in the serum to CSF ratio between baseline and this time point as a group ($p = 0.25$; CI, −43 [−79.5, −6.2]). However, 3 of these 7 ratios changed to > 64 (retest ratios of 800, 1600, and 3200; Table 2) resulting from dilution differences in both serum and CSF: serum (1 dilution increase) and CSF (2 to 4 dilution decrease).

4 | Discussion

We demonstrated that equine *S. neurona* and *N. hughesi* antibodies present in both serum and CSF generally remained within an acceptable variation of 2 dilutions for the different time points when samples were stored at −80°C for up to 24 and 18 months for *S. neurona* and *N. hughesi*, respectively. However, as freezing storage time increased, antibodies in serum and especially CSF showed increasing variability, with antibodies in CSF becoming undetectable in 26% of the samples upon retesting from baseline titers ranging from 5 to 80. Changes in antibody titers in both serum and CSF resulted in alterations of the calculated serum to CSF ratio in 41% ($N = 19/46$) of the paired samples. Of these 19, 16 calculated ratios changed to > 64 and 3 changed to ≤ 64. The clinical relevance of our findings consists of the identification of alterations in the interpretation of results based on calculated ratios from antibody titers with acceptable variability despite the lack of significant difference. On an individual basis, these results have substantial clinical implications and for research purposes. Therefore, awareness of this finding is essential and

raises the question of whether long-term frozen samples should be used for future studies or clinical follow-up.

Structural changes in proteins can occur as a result of extreme temperatures [21]. Protein unfolding is a well-known phenomenon caused by heating a protein solution to high temperatures, which is known as thermal denaturation [21]. On the other hand, when the protein is cooled down from room temperature to lower temperatures, unfolding is referred to as cold denaturation [22], a process in which proteins lose their structure because of an effect known as the hydrophobic effect that can occur at temperatures below −20°C [23]. This raises the question of whether storage at −20°C might be better for protein preservation over time and warrants investigation. When a protein experiences this structural instability, it also loses its functionality [24]. This effect might have occurred in our study and could explain the decrease in antibody titers after long-term storage at −80°C. Freezing and thawing also can play a role in the stability of proteins [16, 21]. However, in our study, samples were thawed only once for retesting.

A previous study evaluating the stability of antibodies for kidney disease in humans reported antibody stability in serum when serum is stored sealed with paraffin paper but not when stored unsealed for more than 7 months [15]. Sealing the samples prevents evaporation and subsequent concentration of antibodies, which can falsely lead to higher antibody titers [15]. Because we observed an increase in antibody titers in some of the samples from our study, an evaporation effect also could have contributed to this apparent increase because of concentration of antibodies. Although we used screw top cryotubes to protect the samples, these were not sealed as described above.

The changes in antibody titers found in our study could lead to misinterpretation of the estimated probability of having disease. In addition to storage factors, test variability also was considered as a possible source for changes in antibody titers. However, these changes were within the accepted variation of 2 dilutions, which were not significantly different for the time points of the study. Nonetheless, these variations in antibodies had an effect on the calculated ratios and interpretation of results. Currently, no definitive antemortem tests are available to diagnose EPM, with the gold standard being histological evidence of the protozoa in the central nervous system [3]. Therefore, the presumptive antemortem clinical diagnosis depends on the detection of antibody titers for *S. neurona* and *N. hughesi* in serum and CSF with calculated ratios based on validated tests such as IFAT and SnSAG 2/4/3 ELISA in horses with neurologic disease [3]. For the IFAT, the higher the antibody titer, the higher the probability of having the disease considering the geographical prevalence of disease (www.vetmed.ucdavis.edu/clinical-laboratory; 2024) [25]. In one report from California, the estimated probability of having EPM was 92% for *S. neurona* and 100% for *N. hughesi* if the antibody titer is ≥1:5 in CSF. These probabilities can be affected by seroprevalence, and therefore might vary depending on geographical location (www.vetmed.ucdavis.eud/clinical-laboratory; 2024) [25].

A notable finding of our study was that 26% (12 of 46) of CSF samples became negative (undetectable antibodies) as storage time increased. Antibodies were not detected in 1 of 9 (11%) CSF samples for the 6–12 months group for *S. neurona*, 7 of 18 (39%) in the 13–18 months group for *S. neurona* and *N. hughesi*, and 4 of 7 (57%) in the 19–24 months group for *N. hughesi*. Differences in CSF results for *S. neurona* vs. *N. hughesi* cannot be compared because the sample size for *N. hughesi* CSF samples was much lower. However, it is possible that antibody stability might be different for *S. neurona* as compared with *N. hughesi*. Previous studies have reported that the addition of protein as a preservative or high protein concentrations in CSF (such as may occur in cases with meningitis) leads to slower degradation of antibodies as well as to a higher preservation of other biological components in CSF [23, 26, 27]. However, the addition of protein to CSF for preservation purposes could preclude the accurate detection of antibodies in some cases. The low protein concentration characteristic of CSF might have contributed to the instability of CSF components resulting in antibody degradation in our samples, but this possibility needs further investigation.

The variability observed in the antibody titers in serum and CSF resulted in changes in serum to CSF ratios, which affect the estimation of probability of EPM as a potential clinical diagnosis. Ratios ≤64 are considered supportive of EPM because of increased intrathecal production of antibodies compared with serum titers [3]. In our study, ratios were seen to decrease or increase depending on changes in IFAT results for serum, CSF, or both. In some cases, horses previously diagnosed with EPM could have been determined as less likely to have the disease based on ratios calculated from stored samples (Table 2). Although 1 to 2 dilutions have been considered an acceptable difference between samples for the interpretation of results [20], these differences had a noticeable effect on the calculation of serum to CSF ratios in our study. In our study, differences in antibody titers could have been attributed to protein degradation (decreasing antibodies) over time, protein instability (samples with low protein content such as CSF), sample evaporation (increasing antibodies), and variability of the test.

Limitations of our study included that the numbers of samples tested were not homogenous across groups. For example, the number of samples per group for *N. hughesi* was underpowered for the 6–12 and 13–18 months freezing storage time point groups. Therefore, further statistical analysis was not possible. Furthermore, the investigation of antibody titers for our study was conducted in two different laboratories within the same institution. Both laboratories are accredited, reputable, use the same validated IFAT protocols, and have long-term highly experienced personnel performing and interpreting the test. Baseline samples were analyzed at the veterinary hospital diagnostic laboratory using the validated protocol developed by the second laboratory. Follow-up samples were analyzed at the research laboratory where the test was developed and validated. Furthermore, antibody titers were not significantly different from baseline compared to those from the first 6 to 12 months of storage at −80°C. This observation suggests that being from different laboratories did not play a major role. For a future study, performing IFAT in replicates by sample and time point in the same laboratory over time would be useful, but was not done in our study because of a lack of sufficient sample materials.

In conclusion, our study indicates that freezing serum and CSF samples at −80°C for more than 18 months might alter *S. neurona*

and *N. hughesi* IFAT results. Therefore, results obtained from long-term frozen samples should be interpreted with caution. Furthermore, despite not having significant differences in antibody titers based on the acceptable 2 dilution variation of the test for both pathogens, the calculation of the serum to CSF ratios on an individual basis changed the interpretation of the probability of disease in 22% to 50% of the cases. These results have important diagnostic implications for research purposes. In addition, these results suggest that the stability of other types of proteins used for diagnostic testing, such as biomarkers, should be investigated to ensure the reliability of the results when performing studies with samples stored frozen for long periods of time.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the University of California Davis Institutional Animal Care and Use Committee (IACUC protocol #23833). All authors have adhered to the principles of ethics of the American Veterinary Medical Association (AVMA). Authors declare human ethics approval was not needed.

Conflicts of Interest

Munashe Chigerwe serves as Associate Editor for the Journal of Veterinary Internal Medicine. He was not involved in the review of this manuscript. The other authors declare no conflicts of interest.

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